Evaluation runs: **bcast** (broadcast-consensus-civl)

generated 2026-11-05 from brain-data 0f578bb07 2026-11-05 11:33:06 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	8	16	conforms	0.0	4	1 h, 250 GB	4	16	brain03, 2026-11-01
3	48	1728	conforms	0.1	2	1 h, 250 GB	8	172	brain03, 2026-11-01
4	384	5308416	conforms	22.0	8	1 h, 250 GB	16	2678	brain03, 2026-11-01
5			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

stateless: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	8	16	14	violation	0.0	2	1 h, 250 GB	16	120	brain03, 2026-11-01
3	48	1728	1688	violation	2.9	4	1 h, 250 GB	1693	81123	brain03, 2026-11-01
4	384			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

Not run (stateless). skipped: a smaller size had reached its limit: 5 decide-early bag, 6 decide-early bag, 6 correct bag, 7 decide-early bag, 7 correct bag, 8 correct bag, 8 decide-early bag, 9 correct bag, 9 decide-early bag, 10 decide-early bag, 10 correct bag.

planned but never scheduled (grid points beyond the ladder): 5 bag decide-early, 5 bag correct, 6 bag decide-early, 6 bag correct.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	8	16	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
3	48	1728	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-01
4	384	5308416	conforms	28.4	25	48 h, 250 GB	brain05, 2026-11-02
5			🕓 (48 h)			48 h, 250 GB	brain05, 2026-11-02

stateful: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	8	16	14	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-01
3	48	1728	1688	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-01
4	384	5308416	5292864	violation	29.6	25	48 h, 250 GB	brain05, 2026-11-03
5				(48 h)			48 h, 250 GB	brain05, 2026-11-03

Not run (stateful). left out by the table rules (timeout): 5 decide-early bag, 5 correct bag, 6 bag decide-early, 6 bag correct.

skipped: a smaller size had reached its limit: 6 decide-early bag, 6 correct bag, 7 decide-early bag, 7 correct bag, 8 correct bag, 8 decide-early bag, 9 correct bag, 9 decide-early bag, 10 decide-early bag, 10 correct bag.

planned but never scheduled (grid points beyond the ladder): 7 bag decide-early, 7 decide-early bag, 7 bag correct, 7 correct bag, 8 correct bag, 8 decide-early bag, 9 correct bag, 9 decide-early bag, 10 decide-early bag, 10 correct bag, 11 correct bag, 11 decide-early bag, 12 decide-early bag, 12 correct bag.

left out by the table rules (invalid: foreign load): 8 bag correct, 8 bag decide-early.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	16	conforms	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	48	1728	conforms	0.1	4	1 h, 250 GB	0.0	0.1	0.0	merge	brain05, 2026-11-03
4	384	5308416	conforms	29.6	23	1 h, 250 GB	0.1	29.6	0.0	merge	brain05, 2026-11-03

offline: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	16	14	violation	0.0	4	1 h, 250 GB	0.0	0.0	0.0	merge	brain05, 2026-11-03
3	48	1728	211	violation	0.1	4	1 h, 250 GB	0.0	0.1	0.0	merge	brain05, 2026-11-03
4	384	5308416	4084	violation	28.7	25	1 h, 250 GB	0.1	28.5	0.2	merge	brain05, 2026-11-03

Not run (offline). skipped: a smaller size had reached its limit: 5 bag correct, 5 decide-early bag, 5 bag decide-early, 5 correct bag, 6 decide-early bag, 6 correct bag, 7 decide-early bag, 7 correct bag, 8 correct bag, 8 decide-early bag, 9 correct bag, 9 decide-early bag, 10 decide-early bag, 10 correct bag.

planned but never scheduled (grid points beyond the ladder): 6 bag decide-early, 6 bag correct.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	16	conforms	3.9	152	10 min, 250 GB	3.9	0.0	0.0	merge	brain01, 2026-11-03
3	48	1728	conforms	4.1	149	10 min, 250 GB	4.1	0.1	0.0	merge	brain01, 2026-11-03
4	384	5308416	conforms	28.9	162	10 min, 250 GB	4.9	28.7	0.2	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	16	14	violation	3.9	152	10 min, 250 GB	3.9	0.0	0.0	merge	brain01, 2026-11-03
3	48	1728	211	violation	4.3	149	10 min, 250 GB	4.1	0.1	0.2	merge	brain01, 2026-11-03
4	384	5308416	4084	violation	27.7	162	10 min, 250 GB	4.9	27.5	0.2	merge	brain01, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 bag decide-early, 5 bag correct, 6 bag decide-early, 6 bag correct.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	8	16	conforms	0.8	150	1 h, 250 GB		brain05, 2026-11-03
3	48	1728	conforms	5.9	447	1 h, 250 GB		brain05, 2026-11-03
4	384		(1 h)			1 h, 250 GB		brain05, 2026-11-03

cross online, Java implementation (JMC) against the Rust specification export: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	8	16	14	violation	0.8	150	1 h, 250 GB	48	brain05, 2026-11-03
3	48	1728	1688	violation	5.0	403	1 h, 250 GB	54	brain05, 2026-11-03
4	384			(1 h)			1 h, 250 GB		brain05, 2026-11-03

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 2 bag correct, 2 bag decide-early, 3 bag decide-early, 3 bag correct, 4 bag decide-early, 4 bag correct, 5 bag correct, 5 bag decide-early, 6 bag correct, 6 bag decide-early.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	8	16	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	48	1728	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02
4	384	5308416	conforms	33.3	23	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	8	16	14	violation	0.0	4	1 h, 250 GB	1	brain01, 2026-11-02
3	48	1728	1688	violation	0.1	4	1 h, 250 GB	2	brain01, 2026-11-02
4	384	5308416	5292864	violation	31.3	27	1 h, 250 GB	2	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 5 bag decide-early, 5 bag correct, 6 bag decide-early, 6 bag correct.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	4.3	166	10 min, 250 GB	4.2	0.0	0.1	merge	brain01, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	1728	1728	conforms	8.0	308	10 min, 250 GB	8.0	0.1	0.0	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	4.1	156	10 min, 250 GB	4.1	0.0	0.0	merge	brain01, 2026-11-03
3	1728	1728	conforms	9.6	381	10 min, 250 GB	9.4	0.1	0.2	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	14	violation	4.2	166	10 min, 250 GB	4.2	0.0	0.0	merge	brain01, 2026-11-03
3	1728	1728	211	violation	8.2	308	10 min, 250 GB	8.0	0.1	0.2	merge	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). refused (an input missing at run time): 4 bag correct, 4 bag decide-early.

planned but never scheduled (grid points beyond the ladder): 5 bag decide-early, 5 bag correct.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	4.2	166	10 min, 250 GB	0.0	4.2	0.0	merge	brain01, 2026-11-03
3	1728	1728	conforms	8.0	308	10 min, 250 GB	0.1	8.0	0.0	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	4.9	156	10 min, 250 GB	0.0	4.1	0.8	merge	brain01, 2026-11-03
3	1728	1728	conforms	9.9	381	10 min, 250 GB	0.1	9.4	0.5	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	14	violation	4.2	156	10 min, 250 GB	0.0	4.1	0.0	merge	brain01, 2026-11-03
3	1728	1728	211	violation	9.4	381	10 min, 250 GB	0.1	9.4	0.0	merge	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). skipped: a smaller size had reached its limit: 4 bag correct, 4 bag decide-early.

planned but never scheduled (grid points beyond the ladder): 5 bag decide-early, 5 bag correct.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	6.1	165	10 min, 250 GB	6.0	0.0	0.1	merge	brain01, 2026-11-03
3	1728	1728	conforms	11.0	2403	10 min, 250 GB	10.9	0.1	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	3.9	152	10 min, 250 GB	3.7	0.0	0.2	merge	brain01, 2026-11-03
3	1728	1728	conforms	8.2	372	10 min, 250 GB	8.2	0.1	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	14	violation	6.2	165	10 min, 250 GB	6.0	0.0	0.2	merge	brain01, 2026-11-03
3	1728	1728	211	violation	10.9	2403	10 min, 250 GB	10.9	0.1	0.1	merge	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). refused (an input missing at run time): 4 bag correct, 4 bag decide-early.

planned but never scheduled (grid points beyond the ladder): 5 bag decide-early, 5 bag correct.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	6.2	165	10 min, 250 GB	0.0	6.0	0.1	merge	brain01, 2026-11-03
3	1728	1728	conforms	11.0	2403	10 min, 250 GB	0.1	10.9	0.1	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	3.7	152	10 min, 250 GB	0.0	3.7	0.0	merge	brain01, 2026-11-03
3	1728	1728	conforms	8.2	372	10 min, 250 GB	0.1	8.2	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	14	violation	4.1	152	10 min, 250 GB	0.0	3.7	0.4	merge	brain01, 2026-11-03
3	1728	1728	211	violation	8.3	372	10 min, 250 GB	0.1	8.2	0.1	merge	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 4 bag correct, 4 bag decide-early.

planned but never scheduled (grid points beyond the ladder): 5 bag decide-early, 5 bag correct.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	16	16	conforms	6.4	167	1 h, 250 GB		brain01, 2026-11-03
3	1728	1728	conforms	12.9	2266	1 h, 250 GB		brain01, 2026-11-03
4	5308416		(1 h)			1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	16	16	conforms	6.1	157	1 h, 250 GB		brain01, 2026-11-03
3	1728	1728	conforms	13.3	2435	1 h, 250 GB		brain01, 2026-11-03
4	5308416		(1 h)			1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	16	16	14	violation	6.5	154	1 h, 250 GB	174	brain01, 2026-11-03
3	1728	1728	1688	violation	12.6	2332	1 h, 250 GB	137	brain01, 2026-11-03
4	5308416			(1 h)			1 h, 250 GB		brain01, 2026-11-02

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 5 bag decide-early, 5 bag correct.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	16	16	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	1728	1728	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	16	16	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	1728	1728	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair decide-early, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	16	16	14	violation	0.0	4	1 h, 250 GB	1	brain01, 2026-11-02
3	1728	1728	1688	violation	0.1	4	1 h, 250 GB	2	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).
skipped: a smaller size had reached its limit: 4 bag correct, 4 bag decide-early.

planned but never scheduled (grid points beyond the ladder): 5 bag decide-early, 5 bag correct.